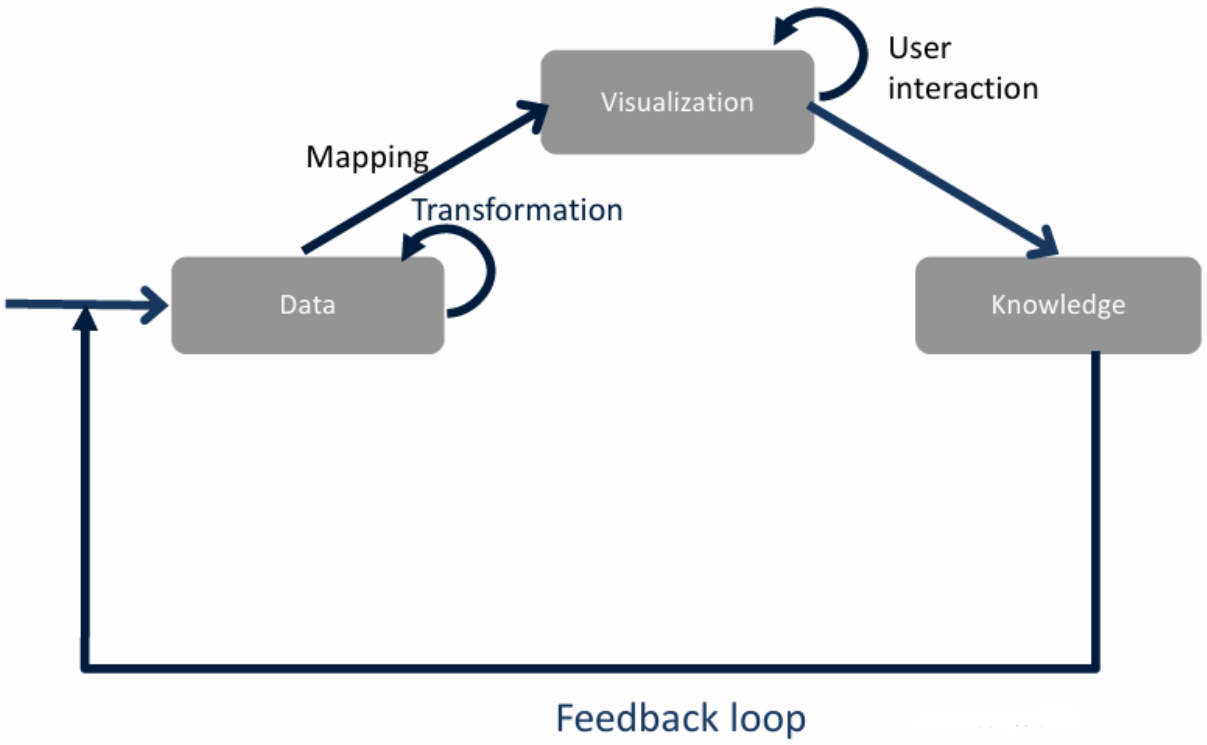
Data Visualization

Notes 2025

Introduction

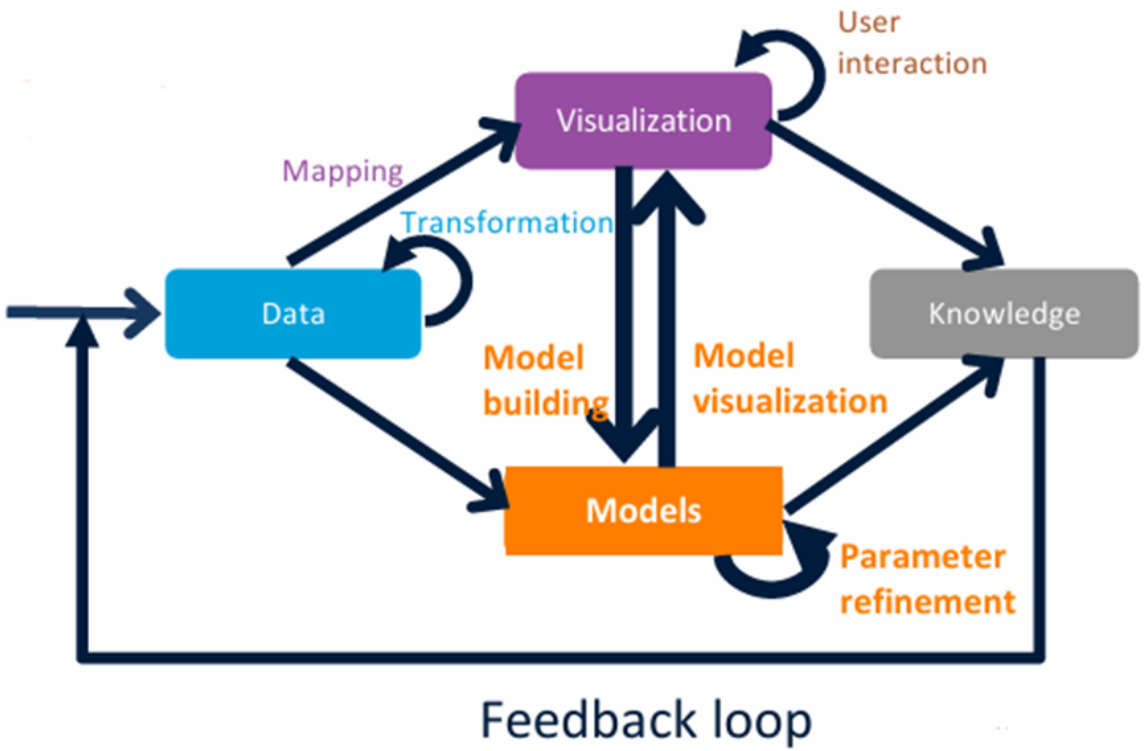
- What is **Data Visualization?** Computer-based visualization systems providing visual representations of datasets, designed to help people carry out tasks more effectively.



Introduction

- What is **Visual Analytics?** Combining automated analysis techniques with interactive visualization for an effective understanding, reasoning and decision making.

1. **Data and Perception**
2. **Visualizations**
3. **Interactivity**
4. **Model Integration**

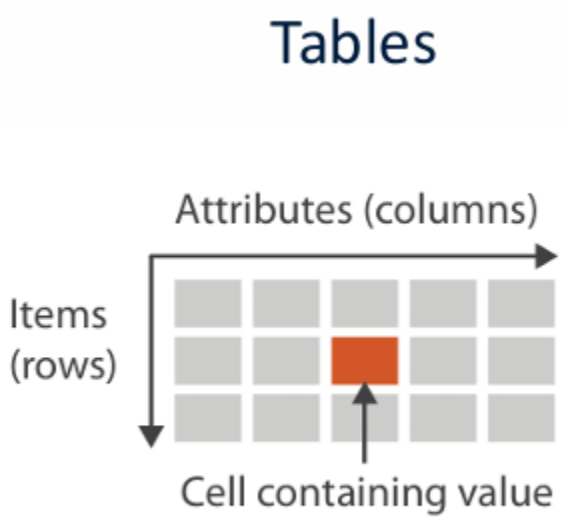


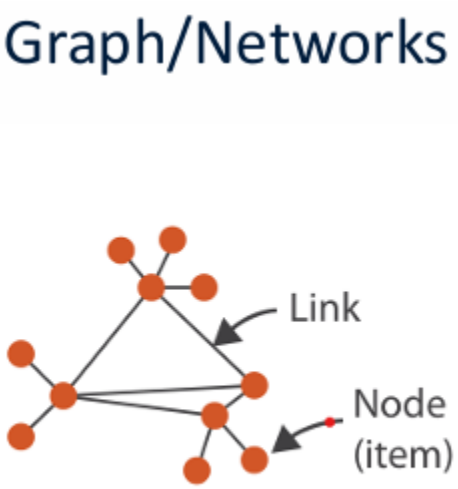
1 – Foundations

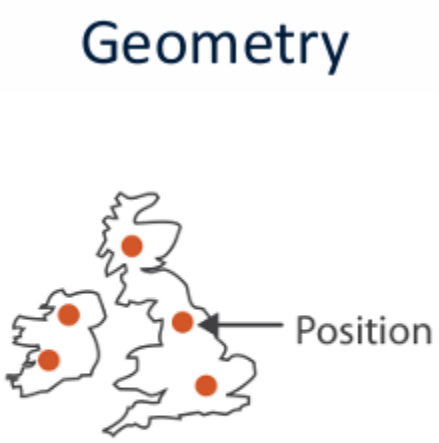
Datasets, Attributes, Tasks, Gestalt laws, Visual channels, Sketching, Color

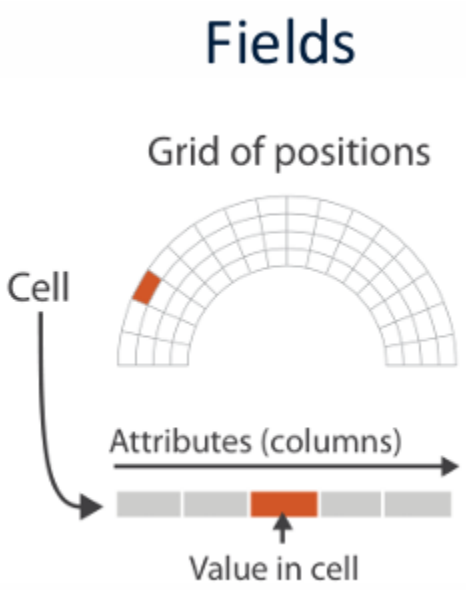
Dataset Types

- **Unstructured Data** (e.g. text, audio) needs to be transformed to a form of structured data before further processing
- **Structured Data** is traditionally in one of the following forms:









Data Types

- **Items**: individual, discrete entities
- **Attributes**: specific properties that can be measured
- **Links**: relationships between items
- **Positions**: spatial information
- **Grids**: strategies for sampling continuous data in terms of geometric and topological relationships between its cells

Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

Data Attribute Types

- **Quantitative**: numeric data with units (+, -, *, /, <, >, =)
- **Ordinal**: data with a semantically relevant order (<, >, =)
  - **Sequential**: homogeneous range from min. to max. value
  - **Diverging**: two sequences that meet at a zero point (e.g. an elevation dataset that contains both mountains and oceans, meeting at 0)
  - **Cyclic**: values wrap around to a starting point (e.g. hours)
- **Categorical**: data that can be used to identify or group (=)
- **None**: data that cannot be used to uniquely identify or group (needs further processing to one of the above)

What?

Datasets

Data Types

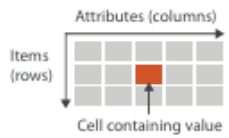
- Items
- Attributes
- Links
- Positions
- Grids

Data and Dataset Types

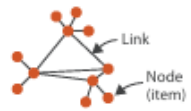
Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

Dataset Types

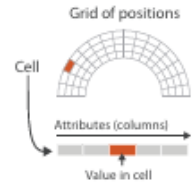
- Tables



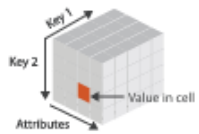
- Networks



- Fields (Continuous)



- *Multidimensional Table*



- *Trees*

- Geometry (Spatial)

Dataset Availability

- Static

- Dynamic

Attributes

Attribute Types

- Categorical

- Ordered
  - *Ordinal*

  - *Quantitative*

Ordering Direction

- Sequential

- Diverging

- Cyclic

What can be visualized

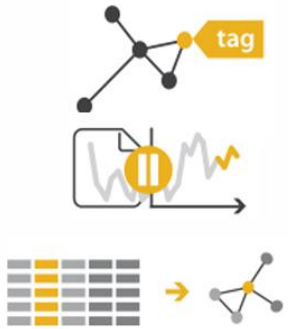
Tasks

- First step of any visualization design: identify the **user needs**: what does the visualization (system) need to accomplish?
- User needs should be described in abstract, *domain-independent* language. This allows for search and comparison of different visualization techniques
- A structured approach is to use *{Action, Target}* pairs:
  - **Action**: goal of the user
  - **Target**: the data features of interest
- **High Level Actions**: for what goal is the visualization applied?
- **Mid-Level Actions**: how is the user searching for the information?
- **Low Level Actions**: how much information needs to be considered at once?

High Level Actions

- **Consume Information** (data is already present)
  - **Explore**: The user should find new knowledge
  - **Present**: The user presents known information to others
  - **Enjoy**: The user should enjoy the visualization

- **Produce Information** (data will be added)
  - **Annotate**: Add attribute to data element
  - **Record**: Save visualization / analysis artifact
  - **Derive**: Produce new data based on existing data



- *E.g. What is the goal? Explore / Present / Enjoy / Annotate / Record / Derive?*

Mid-Level Actions

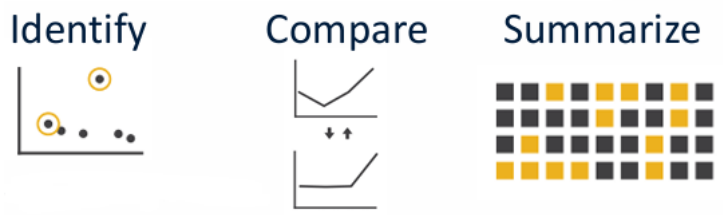
- **Target**: What item are we looking for?
- **Location**: What attributes / identifiers do the items have?

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

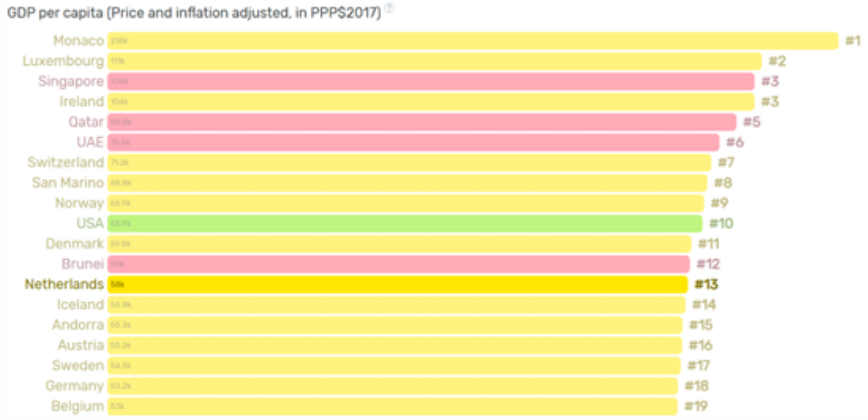
- *Example*:
  - **Lookup** how Maastricht is connected
  - **Browse** alternative routes from Maastricht to Utrecht
  - **Locate** the longest intercity stretch without intermediate stations
  - **Explore** how the Netherlands is connected

Low Level Actions

- What to do with the targets from Mid-Level actions?
  - **Identify** attributes of a single target
  - **Compare** attributes of multiple targets
  - **Summarize** attributes of all targets

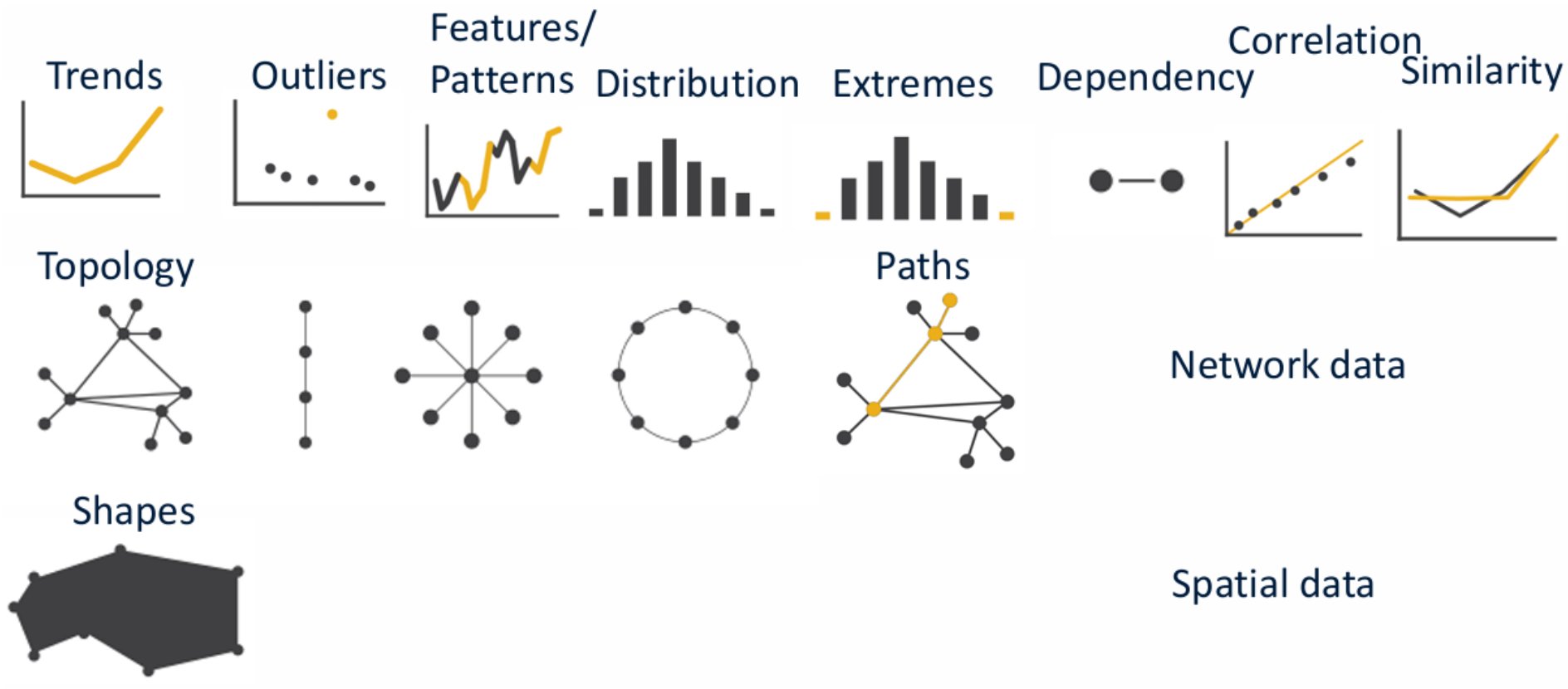


- *Example*:
  - Identify the rank of the Netherlands for GDP per Capita:
    - **Lookup → Identify**
  - Compare the Netherlands and the highest Asian country:
    - **Lookup + Browse → Compare**
  - Summarize general trends in GDP per Capita:
    - **Explore → Summarize**

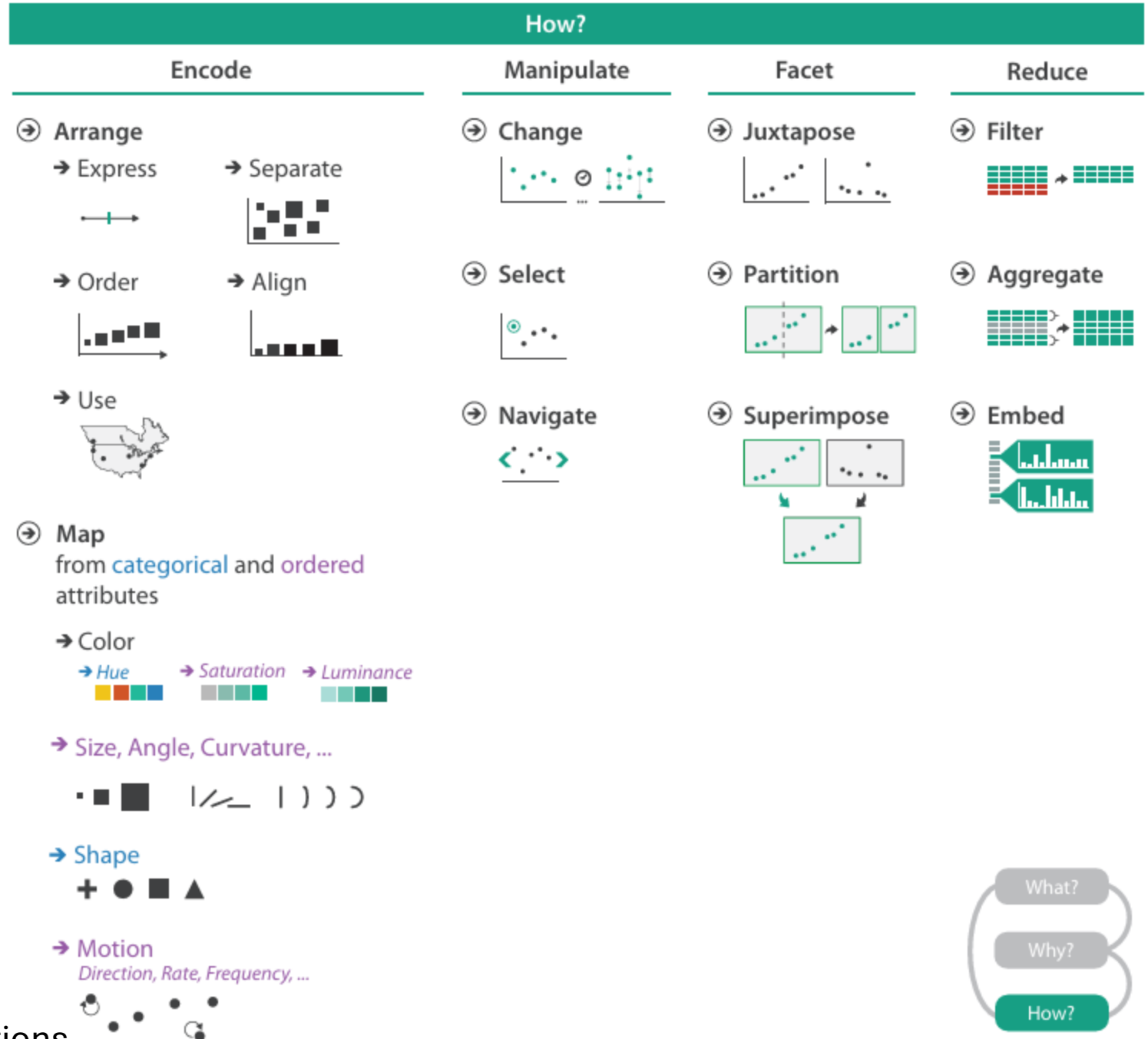


Targets

- What features / attributes can we target?



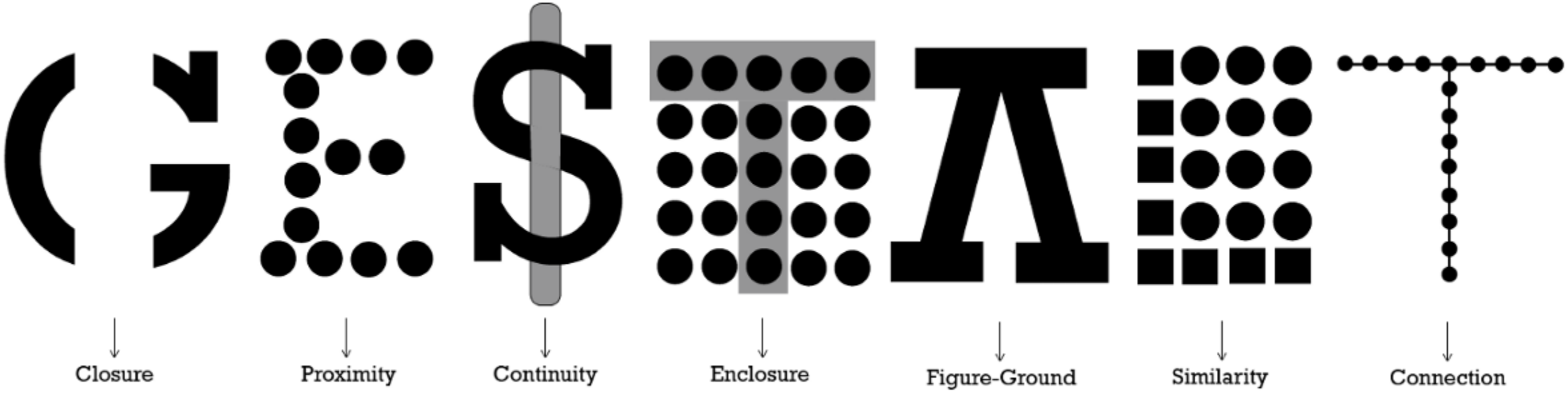
Why people are using visualization



How to design visualizations

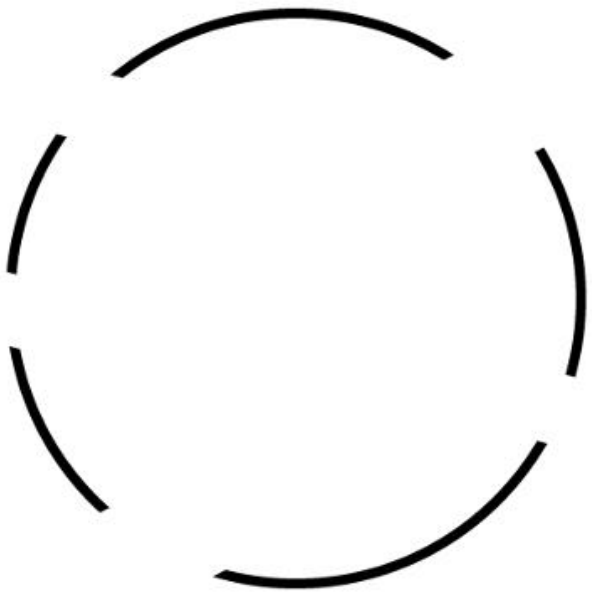
Gestalt Laws of Perception

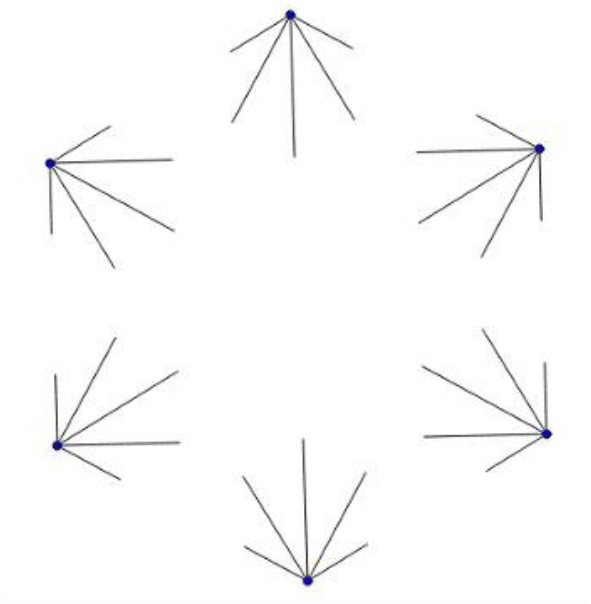
- 7 laws that describe how humans tend to interpret visual information



Closure

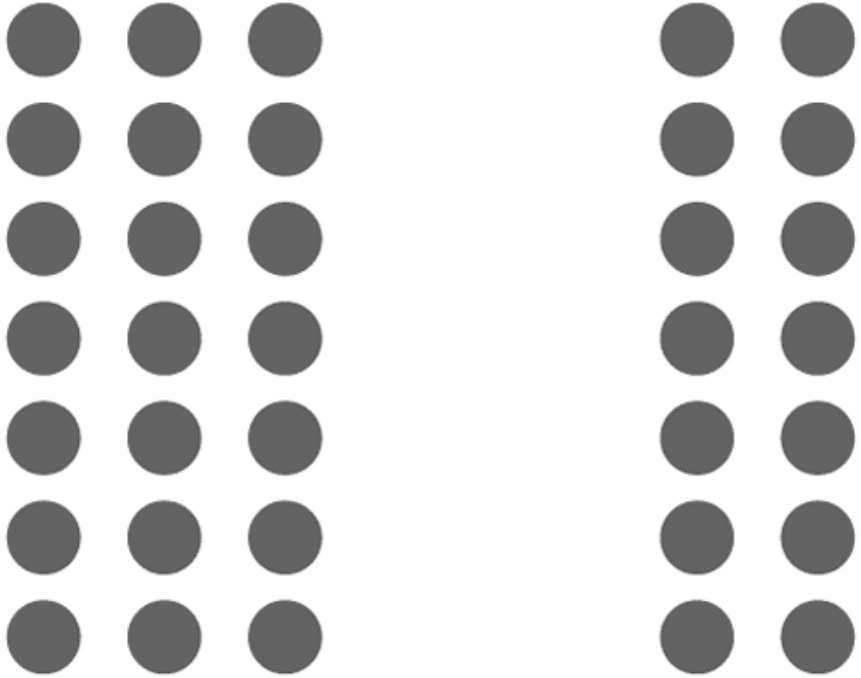
- Our brains perceive incomplete shapes as complete by filling in the gaps

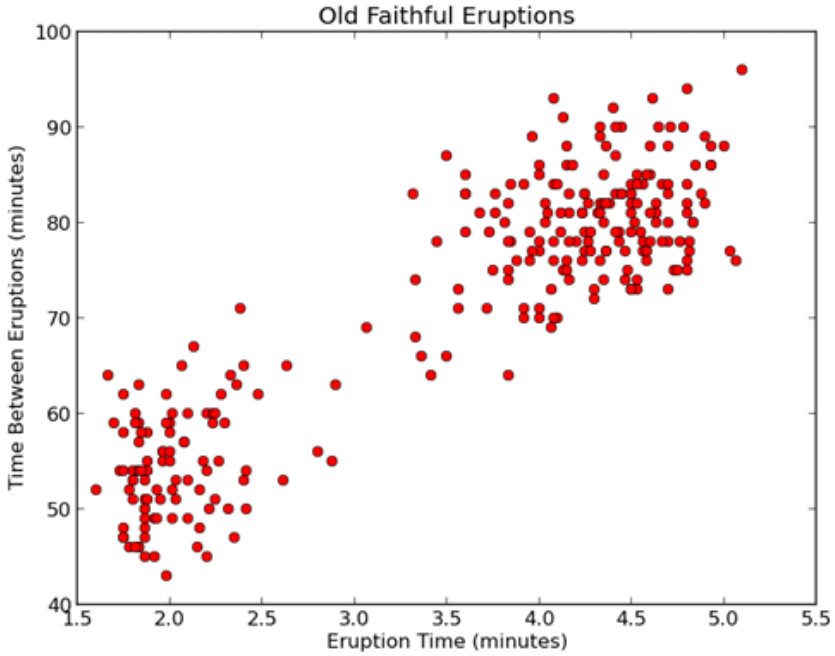




Proximity

- Objects placed near each other are perceived as belonging to a group





Continuity

- The human eye automatically follows the smoothest path

Enclosure

- Objects contained within a common boundary are perceived as a group

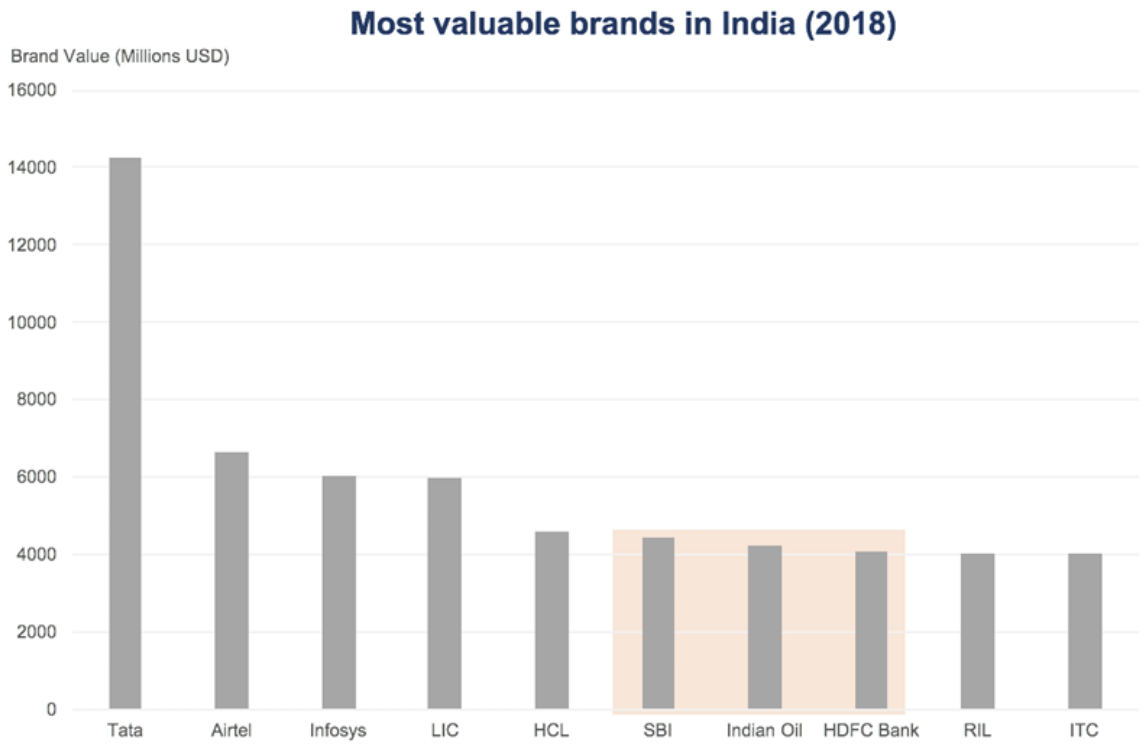


Figure-Ground

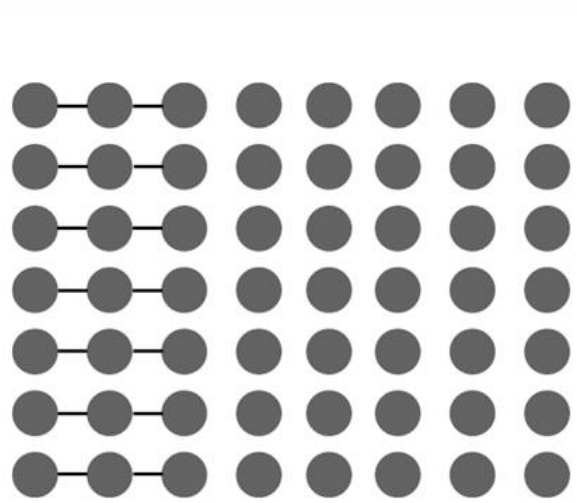
- The human visual system perceives elements as either the focal *“figure”* in the foreground, or the *“ground”* in the background

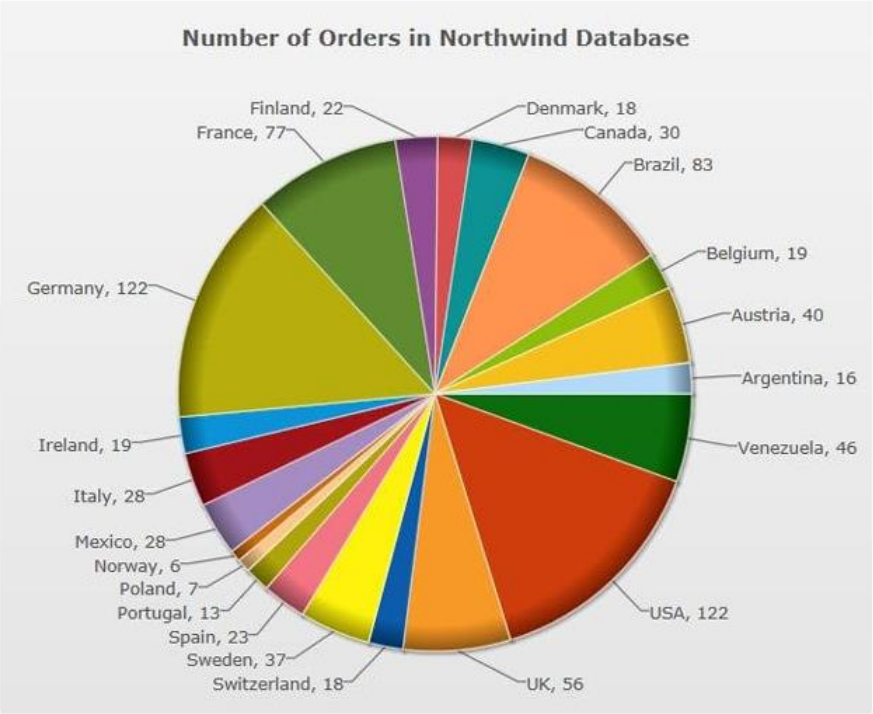
Similarity

- People tend to group visually similar elements together

Connection

- People perceive visually connected elements as belonging to the same group



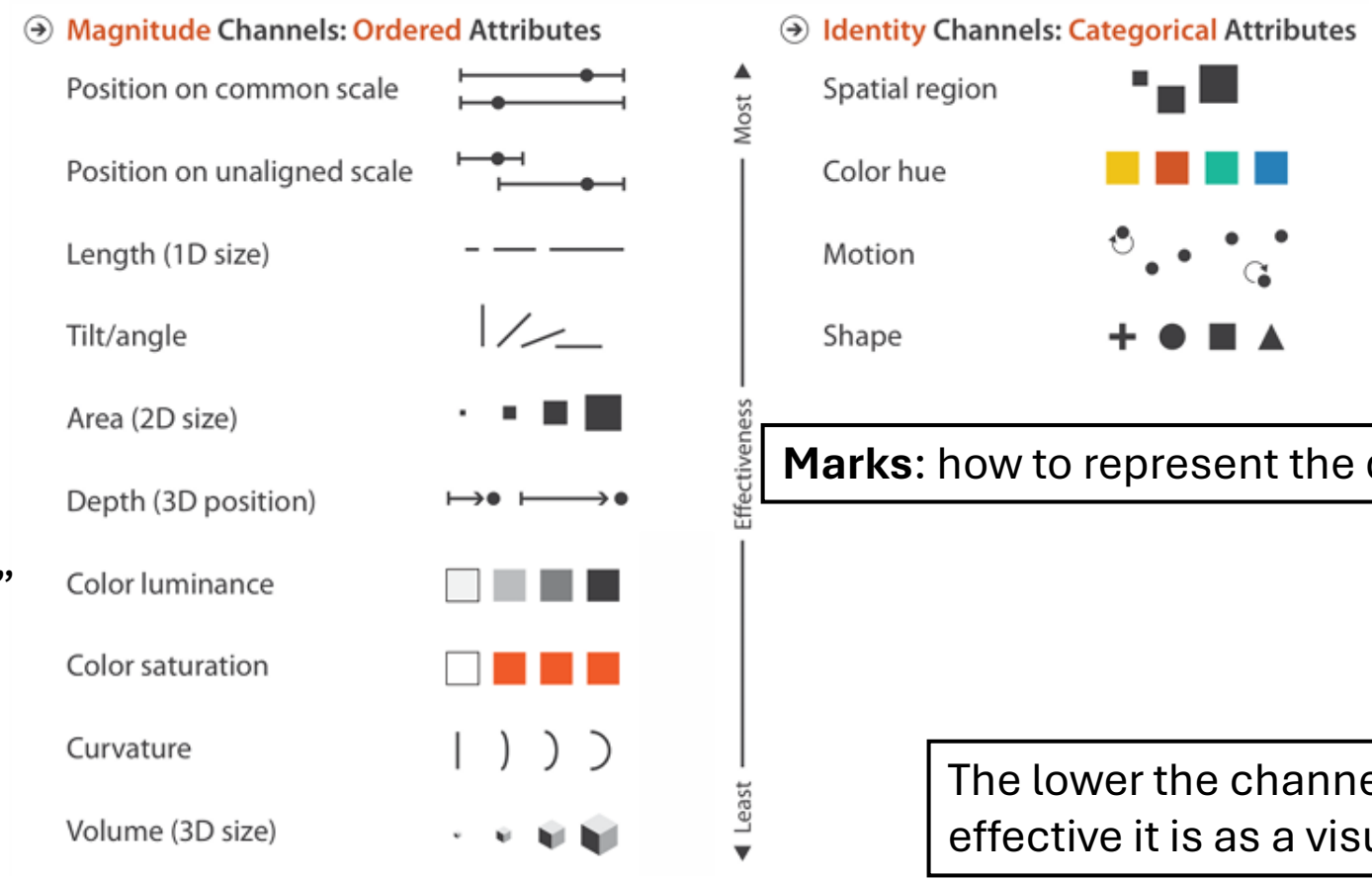


Visual Channels

- **Visual channels** are the foundation of every visualization

e.g. a clock

“which area is bigger?” is more difficult to answer than “which line is longer?”



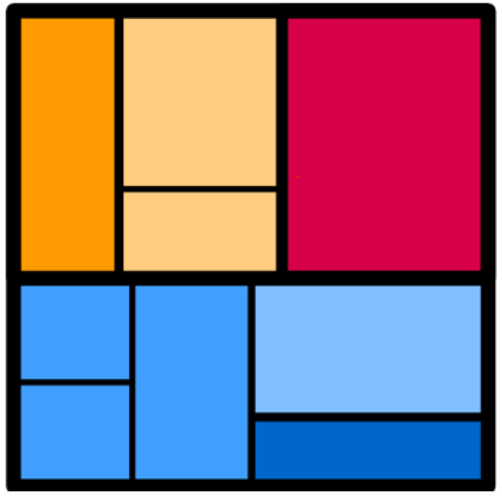
e.g. separate columns for each class

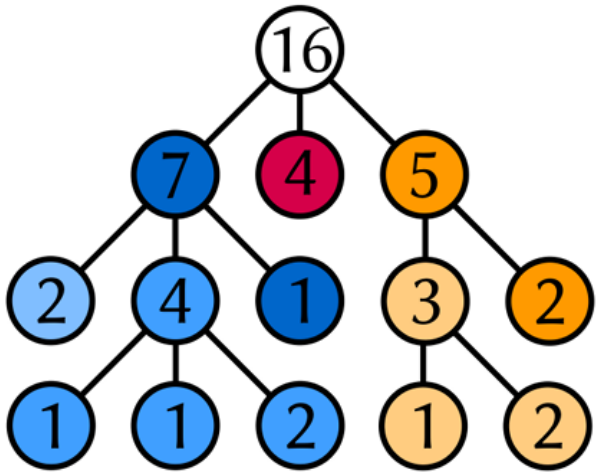
Marks: how to represent the data itself (point, line, area)

The lower the channel, the less effective it is as a visualization technique

Visual Channels

- Example:
  - Child *u* of node *v* → Spatial region of *u*
  - Value → Size
  - First split → Hue
  - Depth → Luminance
  - Depth → Length (line width)
  - Mark → Area



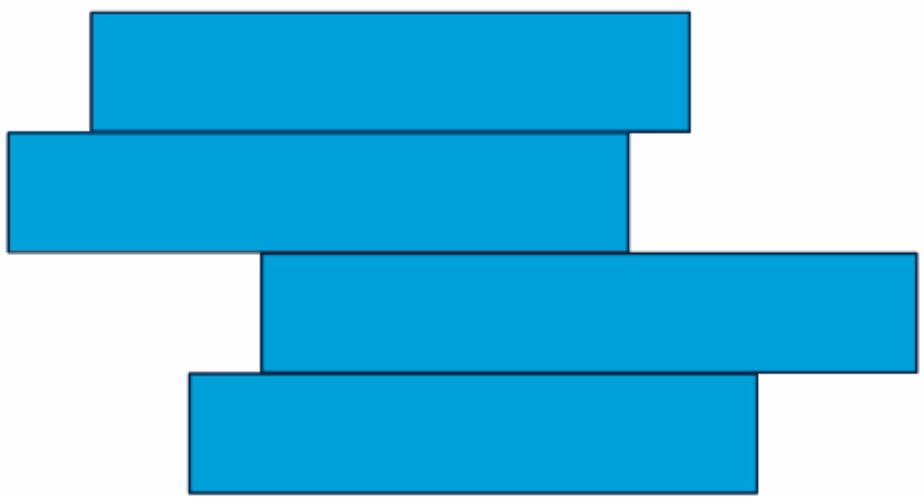


- (Here Depth is a **Double Mapping**, as it's mapped to two visual channels)

Visual Channels

- Visual channels often interfere with another, some more than others:

Length



Length + x-Position

Length + Luminance

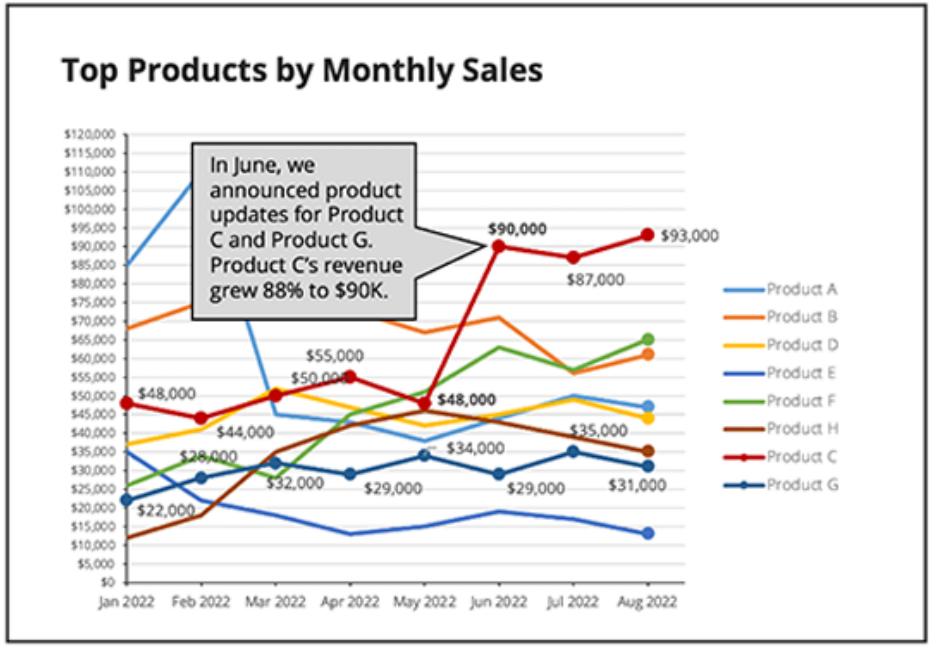
Visual Channels & Gestalt Laws

- Gestalt laws are closely linked to visual channels and marks:
  - **Closure**
  - **Proximity**: Position on (un)aligned scale, Depth, Spatial region
  - **Continuity**
  - **Enclosure**: Spatial region
  - **Figure-Ground**
  - **Similarity**: Everything except Position on aligned scale, Depth, Spatial region
  - **Connection**: Position on (un)aligned scale, Depth, Spatial region, line mark
- Some can't be mapped directly, but can be used to aid information density

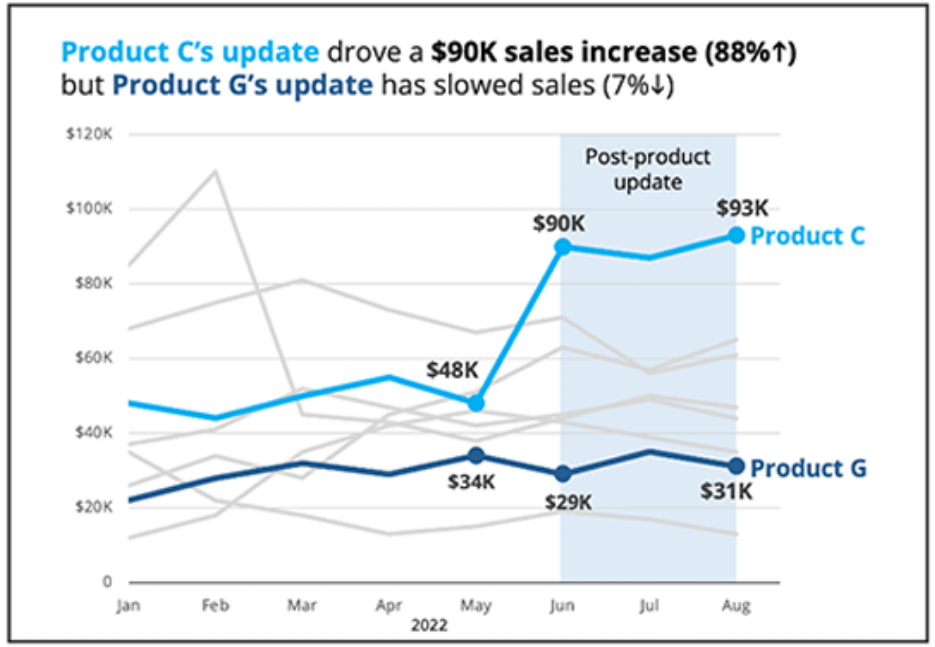
Data-Ink Ratio

- Visualizations should aim to maximize the data-ink ratio
- *Maximize* the meaningful space used, *minimize* extra elements (clutter)

With Clutter



Without Clutter



Sketching

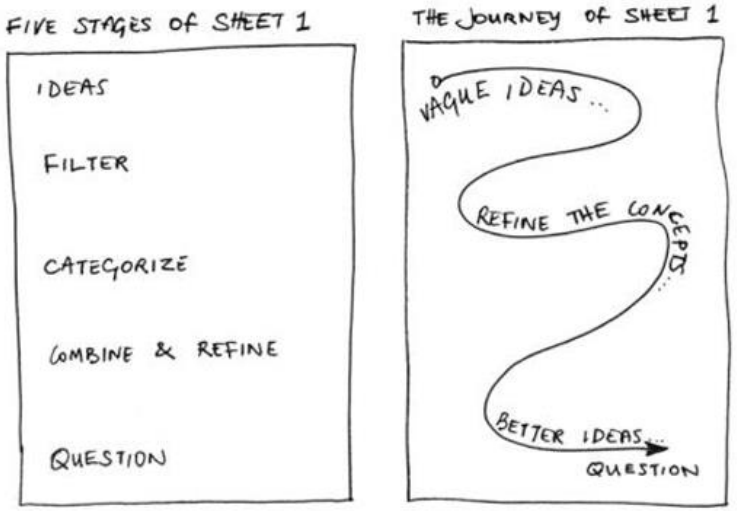
- A sketch is formally defined as *“a rapidly executed freehand drawing that is not usually intended as a finished work”*

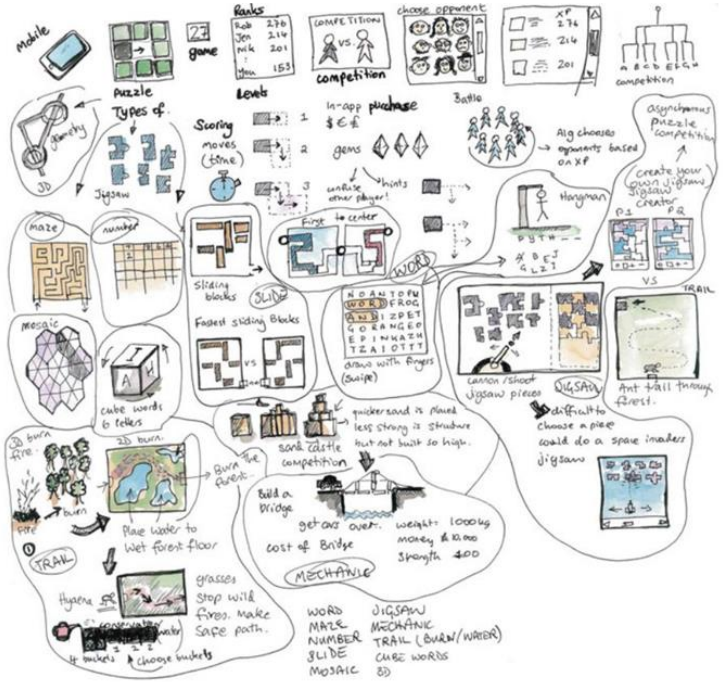
- It is a cheap, fast, flexible medium for exploration and discussion of ideas

- A good sketch:
  - Explains the core idea well (doesn’t have to be a good idea)
  - Stands on its own (idea can be explained using just the sketch)
  - Has an appropriate level of detail (initially low, later sketches can fill in gaps)

5 Design Sheet Methodology

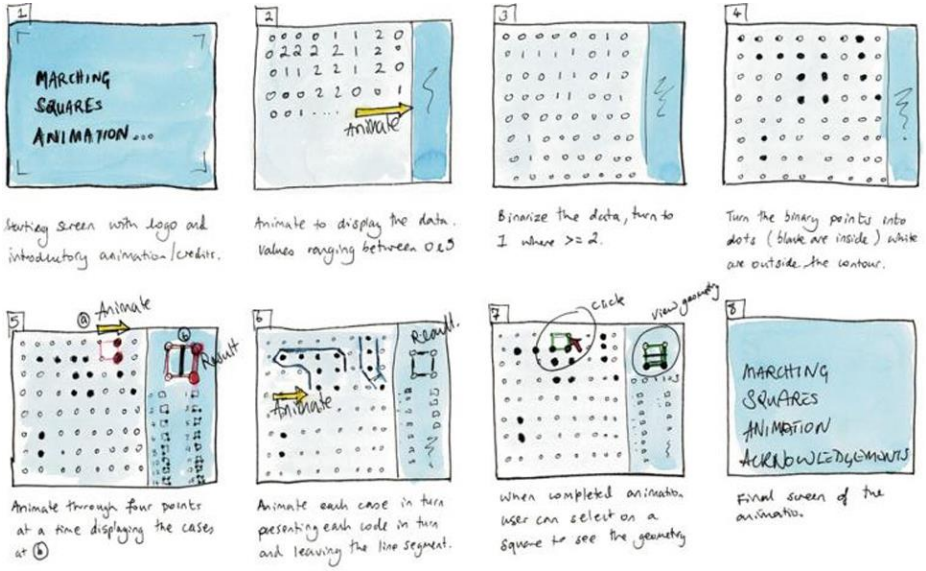
- Sheet 1: **Consideration Space Exploration**, large amount of initial solutions
  1. **Ideas**: produce ideas that *might* solve the problem
  2. **Filter**: consider what may not work in its current design, but *might* work differently
  3. **Categorize**: group similar ideas together
  4. **Combine and Refine**: combine ideas to make them better
  5. **Question**: what ideas have potential?

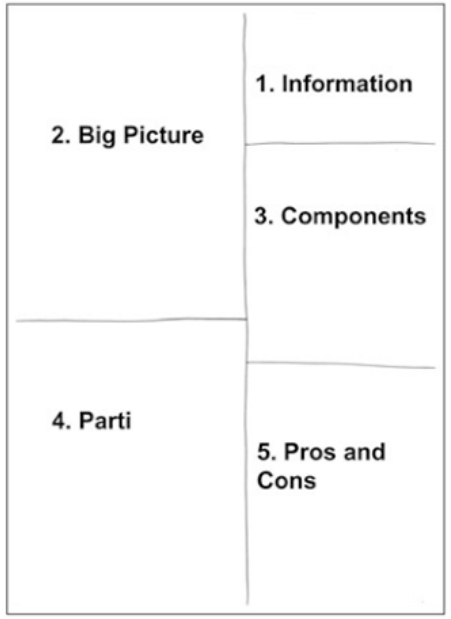




5 Design Sheet Methodology

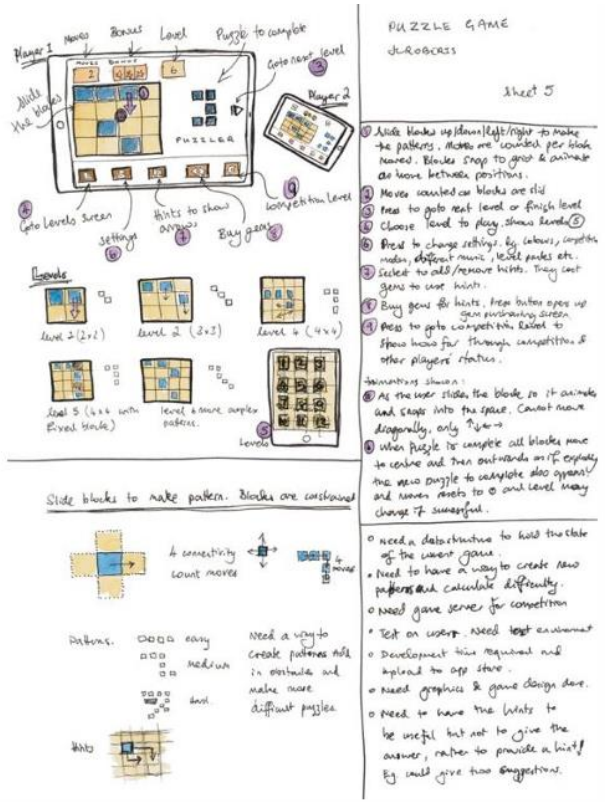
- Sheet 2, 3, 4: **Proposal Space Exploration**, 3 main proposed solutions
  - **Information**: author, date, sheet number, title, etc.
  - **Big Picture**: more detailed sketches that describe the core idea
  - **Components**: the key interactions in the design
  - **Parti**: what is unique about the design? (compared to the other ideas)
  - **Pros and Cons**: what are the advantages and disadvantages of this design?





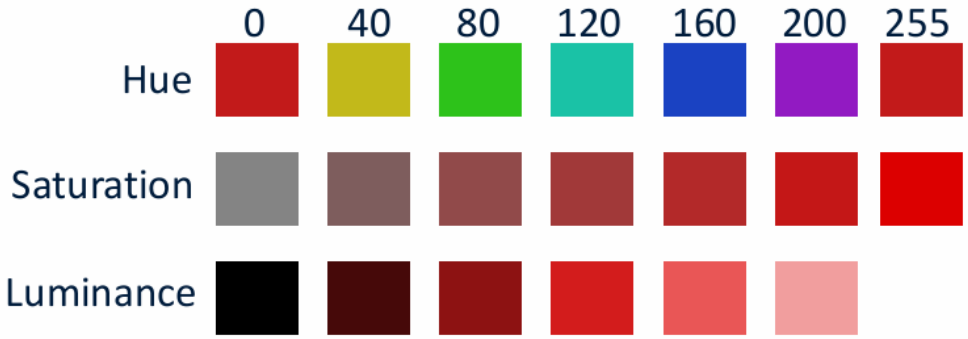
5 Design Sheet Methodology

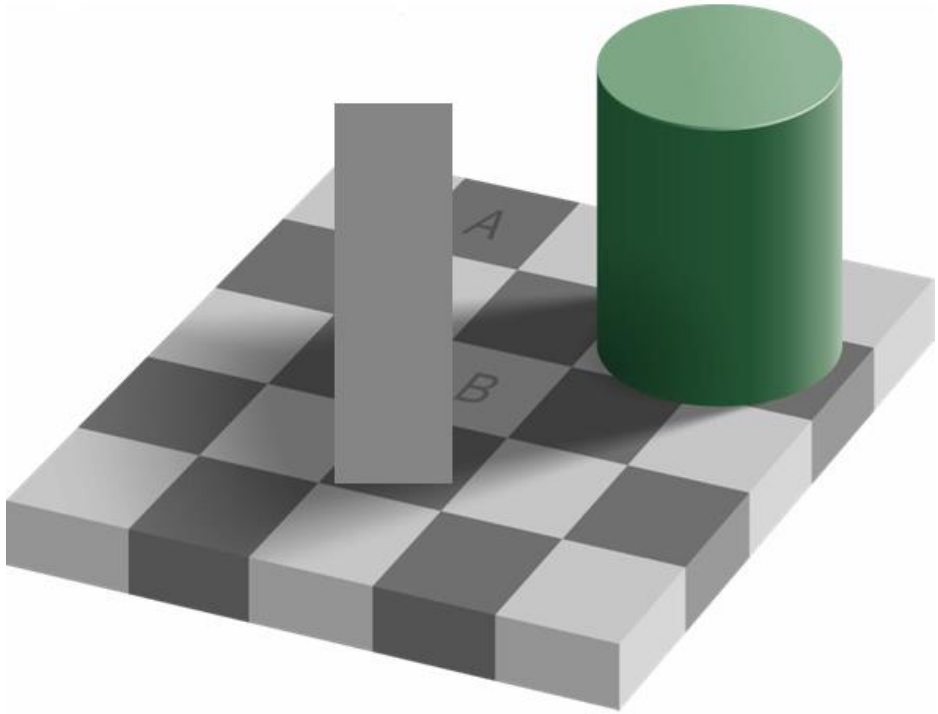
- Sheet 5: Realization of the design that will be implemented. Should be so detailed that you can give it someone else, and they can make your vision
- How to decide:
  - **Goals**: does it achieve my original goal?
  - **Fit**: does it fit your user?
  - **Usable**: is it clear how to use it?
  - **Useful**: will people use it?
  - **Aesthetics**: is it pretty enough?
  - **Cost**: is this realizable?
  - **Sustainable**: how will this be used in the future?
  - **Unique**: do similar things already exist?
  - **Buildable**: do I have the capacity to make this?



Color

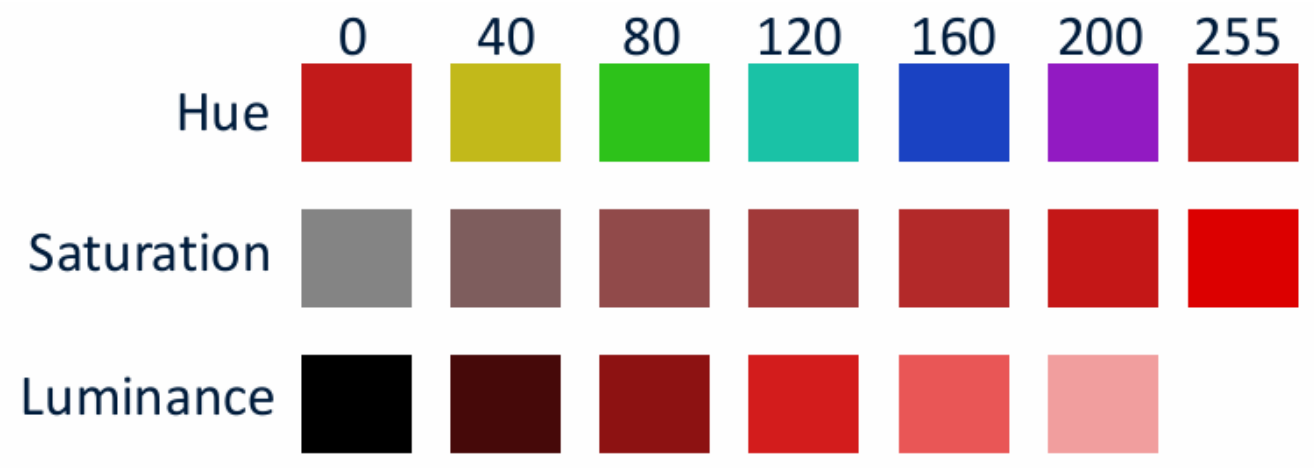
- Color conflicts with other visual perceptions very easily (this is why color luminance places quite low on the visual channels scale)
- Interpolation in RGB is difficult: we see combinations of wavelengths, not RGB
- **HSL** (Hue, Saturation, Luminance) more closely matches the way humans observe colors

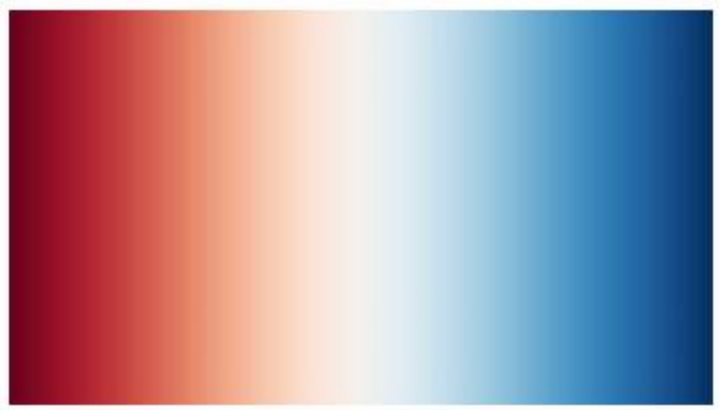




Color Scales

- Categorical Data → **Categorical Color Scale** (go through the hues)
- Ordinal Data → **Sequential Color Scale** (go through saturation or luminance)
- Quantitative Data → **Sequential Color Scale** (saturation or luminance)
- Data with a semantically relevant zero-point (money, elevation, temperature, etc.) → **Diverging Color Scale** (luminance through two hues)





Color Scales

- It is difficult to compare shades of color to each other:

- Saturation is *not* a linear scale

Color Scales

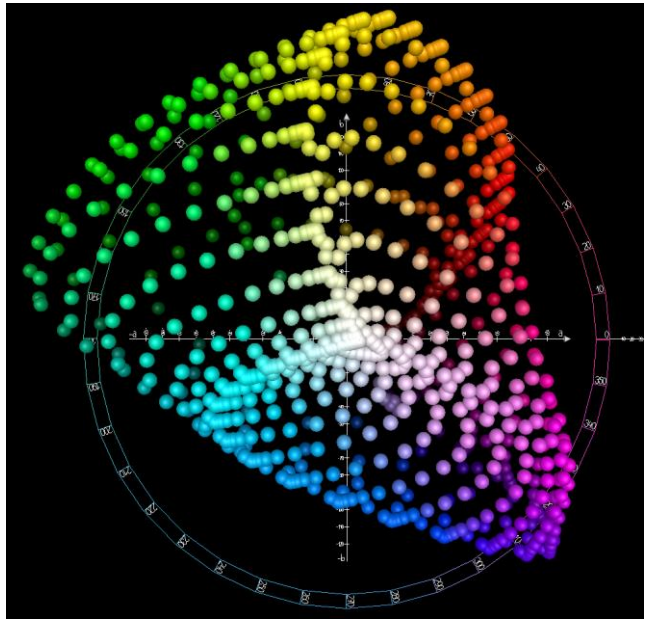
- Luminance is slightly better than saturation (hence it is higher on the visual channel scale), but it is still not linear:

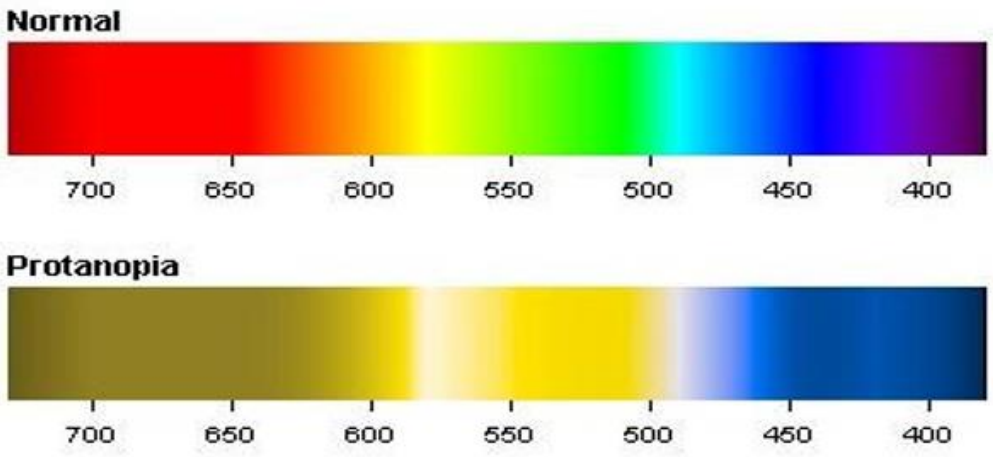
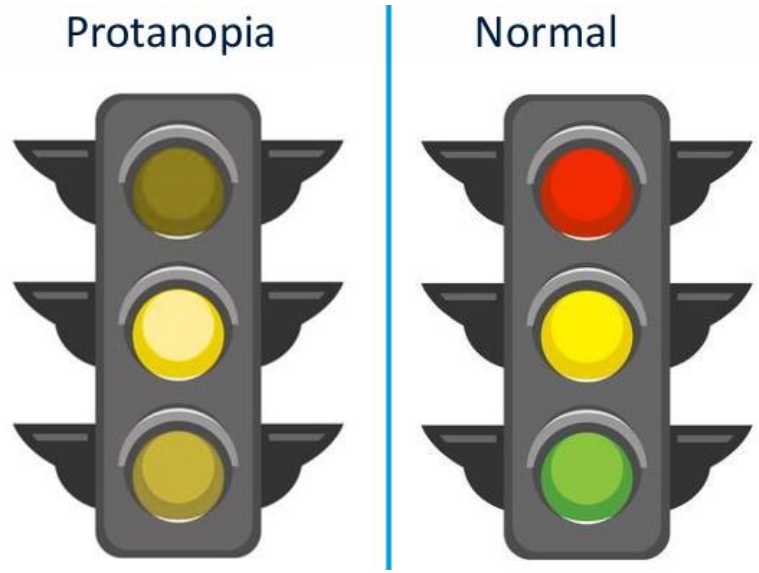
- Colors are great for identification, but it is hard to judge the exact differences

Color Scales

- The CIELAB color space is **Perceptually-Uniform**. It is modelled after how the brain interprets color and uses non-linearity to approximate uniformity
- The distance between colors is denoted as ΔE, with $\Delta E = 1$ meaning that the probability of someone correctly distinguishing between the colors is 50%

$\Delta E = 100$ $\Delta E = 30$ $\Delta E = 3.5$



Colorblindness

- Roughly 9% of the population is red-green colorblind
- Ways to help colorblind people understand your visualizations:
  - Double Encoding (e.g. traffic light has color + y-position)
  - Avoid red + green and green + yellow (red + blue is always usable)
  - Scale through saturation or luminance, not hue
  - Allow users to choose their own visual variable mapping